A Physical Introduction to Higher Mathematics

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Contents

Part I

Waves

1 Linear Structure and Spectra

Definition 1.1 (Vector spaces and linear maps)**.** Let k be a field. A *k-vector space* is an abelian group V with an action $k \times V \to V$ satisfying distributivity, associativity, and $1v = v$. A map $T : V \to W$ is *linear* if $T(av + bw) = aT(v) + bT(w)$.

Definition 1.2 (Subspaces, kernels, images, and quotients)**.** A *subspace* of V is a subset closed under linear combinations. For a linear map $T : V \to W$, define

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For a subspace $U \subseteq V$, the *quotient* V/U is the vector space of cosets $v + U$. The *direct sum* $V \oplus W$ is $V \times W$ with componentwise operations.

Exercise 1.1 (Universal linear constructions)**.** Prove that $\ker T$ and $\operatorname{im} T$ are subspaces. State and prove the universal properties of V/U and $V \oplus W$. Deduce $V/\ker T \cong \operatorname{im} T$ and the three isomorphism theorems.

Definition 1.3 (Duals and pairings)**.** The *dual* of V is $V^* = \operatorname{Hom}_k(V, k)$. A *pairing* of V and W is a bilinear map $V \times W \to k$. It is *nondegenerate* if each nonzero argument pairs nontrivially with some argument in the other space.

Definition 1.4 (Tensor products)**.** A *tensor product* of V and W is a vector space $V \otimes W$ with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $b : V \times W \to X$ factors uniquely through a linear map $\tilde{b} : V \otimes W \to X$. Tensor powers and multilinear maps are defined by iteration.

Exercise 1.2 (Duality and multilinearity)**.** For finite-dimensional spaces, construct the natural maps

$$V^* \otimes W \longrightarrow \operatorname{Hom}(V, W), \qquad V^* \otimes W^* \longrightarrow (V \otimes W)^*.$$

Prove that they are isomorphisms, identify the evaluation pairing, and prove the tensor–hom adjunction $\operatorname{Hom}(U \otimes V, W) \cong \operatorname{Hom}(U, \operatorname{Hom}(V, W))$.

Definition 1.5 (Inner products and orthogonality)**.** An *inner product* on a real vector space is a positive-definite symmetric bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate-linear in the first variable. Write $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors are *orthogonal* when their inner product is zero, and $U^\perp = \{v : \langle u, v\rangle = 0 \text{ for every } u \in U\}$.

Definition 1.6 (Projections and adjoints)**.** A *projection* is an endomorphism P with $P^2 = P$. An *orthogonal projection* is a projection with $\ker P = (\operatorname{im} P)^\perp$. The *adjoint* of $T : V \to W$ is a map $T^* : W \to V$ satisfying $\langle Tv, w\rangle = \langle v, T^*w\rangle$.

Exercise 1.3 (Orthogonal decomposition and best approximation)**.** Let V be finite-dimensional and $U \subseteq V$. Prove $V = U \oplus U^\perp$, construct the orthogonal projection P_U, and prove that $P_U v$ is the unique element of U minimizing $\|v - u\|$. Prove that P is an orthogonal projection if and only if $P = P^* = P^2$.

Definition 1.7 (Isometries and spectral classes)**.** A linear map is *orthogonal* over $\mathbb{R}$, or *unitary* over $\mathbb{C}$, if it preserves the inner product. An endomorphism A is *self-adjoint* if $A = A^*$ and *normal* if $AA^* = A^*A$. A scalar λ is an *eigenvalue* if $\ker(A - \lambda\,\mathrm{id}) \neq 0$; a subspace U is *invariant* if $A(U) \subseteq U$.

Definition 1.8 (Spectral decomposition)**.** A *spectral decomposition* of an endomorphism A is an expression

$$A = \sum_{\lambda \in \mathrm{spec}(A)} \lambda P_\lambda, \qquad P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda, \qquad \sum_\lambda P_\lambda = \mathrm{id},$$

with orthogonal projections P_λ.

Exercise 1.4 (Finite-dimensional spectral theorem)**.** Prove that adjoints exist uniquely. Prove that a complex endomorphism is unitarily diagonalizable if and only if it is normal. Deduce spectral decompositions for self-adjoint, unitary, and normal operators, including the reality of self-adjoint spectra and the unit modulus of unitary spectra.

Exercise 1.5 (Diagonalization and invariant subspaces)**.** For an endomorphism with split characteristic polynomial, prove that it is diagonalizable if and only if the space is the direct sum of its eigenspaces. Show that normality removes generalized eigenvectors. Prove that commuting normal endomorphisms admit a common orthogonal spectral decomposition.

2 Oscillation and Fourier Analysis

Definition 2.1 (Linear oscillator)**.** A *linear oscillator* on a finite-dimensional real inner-product space V is an equation

$$M\ddot{q} + Kq = 0,$$

where M is positive definite and K is self-adjoint and positive semidefinite. A *normal mode* is a nonzero $v \in V$ for which $q(t) = a(t)v$ solves the equation for some nonconstant a.

Exercise 2.1 (The harmonic oscillator)**.** Solve $m\ddot{q} + kq = 0$ for $m, k > 0$. Prove conservation of $E = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$, determine the period, and deduce from the measured period of a small-amplitude mass–spring system the ratio k/m.

Exercise 2.2 (Coupled oscillators and normal modes)**.** Transform $M\ddot{q}+Kq = 0$ by $M^{1/2}$. Apply the spectral theorem to prove the existence of an M-orthogonal basis of normal modes and derive their frequencies. For a fixed-end chain of N equal masses joined by equal springs, compute the frequencies and mode shapes and identify the modes seen in a Chladni-type standing-wave measurement.

Definition 2.2 (Vibrating string)**.** For length L, tension τ, and linear density ρ, a *vibrating string* is a function $u : [0, L] \times \mathbb{R} \to \mathbb{R}$ satisfying fixed-end conditions $u(0, t) = u(L, t) = 0$ and

$$\rho\, \partial_t^2 u = \tau\, \partial_x^2 u.$$

Exercise 2.3 (String modes and measured pitch)**.** Derive the vibrating-string equation from transverse force balance in the small-slope limit. Determine all separated normal modes and their frequencies. Prove that the fundamental frequency is $f_1 = (2L)^{-1}\sqrt{\tau/\rho}$ and compare the predicted ratios $f_n/f_1 = n$ with the harmonics of a stretched string.

Definition 2.3 (Orthogonal expansions)**.** Let H be an inner-product space. For an orthogonal family (e_n), the *orthogonal coefficients* of f are $\widehat{f}(n) = \langle e_n, f\rangle/\|e_n\|^2$. A *Fourier series* on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ is the formal expansion $f \sim \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{inx}$, where $\widehat{f}(n) = (2\pi)^{-1}\int_0^{2\pi} f(x)e^{-inx}\,dx$.

Exercise 2.4 (Bessel, completeness, and Parseval)**.** Prove Bessel's inequality for every finite orthogonal family. Prove that an orthonormal family (e_n) is complete precisely when

$$\|f\|^2 = \sum_n |\langle e_n, f\rangle|^2$$

for every f. Prove completeness of $(e^{inx})_{n\in\mathbb{Z}}$ in $L^2(\mathbb{T})$ and obtain Parseval's identity and L^2 convergence of Fourier series.

Exercise 2.5 (Spectral evolution on an interval)**.** Expand suitable fixed-end initial data in string modes and derive the unique solution of the wave equation. Expand suitable initial temperature data in the same spatial eigenfunctions and derive the fixed-end solution of $\partial_t u = \kappa\partial_x^2 u$. Prove conservation of wave energy and monotone decay of heat energy directly from the spectral coefficients.

Definition 2.4 (Fourier transform and convolution)**.** For $f \in L^1(\mathbb{R}^n)$, its *Fourier transform* is

$$\mathcal{F}f(\xi) = (2\pi)^{-n/2}\int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}\,dx.$$

For $f, g \in L^1(\mathbb{R}^n)$, their *convolution* is $(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\,dy$.

Exercise 2.6 (Fourier inversion, convolution, and Plancherel)**.** For Schwartz functions, prove Fourier inversion and $\mathcal{F}(f * g) = (2\pi)^{n/2}(\mathcal{F}f)(\mathcal{F}g)$. Prove that $\mathcal{F}$ preserves the L^2 inner product and extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$. Deduce the Plancherel theorem.

Exercise 2.7 (Wave and heat propagators)**.** Use the Fourier transform to solve the Cauchy problems for

$$\partial_t^2 u = c^2\Delta u, \qquad \partial_t u = \kappa\Delta u$$

on $\mathbb{R}^n$. Identify their spectral-mode evolution, derive the Gaussian heat kernel and its convolution law, and prove finite propagation speed for the one-dimensional wave equation.

Exercise 2.8 (Interference and Fourier optics)**.** For coherent scalar waves with complex amplitudes a_1, a_2, derive $|a_1 + a_2|^2$ and the fringe spacing in Young's double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

Part II
Symmetry

3 Groups and Representations

Definition 3.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Exercise 3.1 (The group isomorphism theorems)**.** Prove that kernels are normal and that a normal subgroup is the kernel of the quotient homomorphism. Prove the three isomorphism theorems for groups and the universal property of G/N.

Definition 3.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Exercise 3.2 (Orbit–stabilizer and class equation)**.** Construct a natural bijection $G/G_x \to Gx$. For finite G, deduce the orbit–stabilizer formula and the class equation from the conjugation action. Use these results to prove that every group of prime order is cyclic.

Definition 3.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 3.3 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 3.4 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 3.4 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

Exercise 3.5 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form an orthonormal basis of the class functions and that $\langle \chi_U, \chi_V \rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Exercise 3.6 (Symmetry projections)**.** For an irreducible character χ of a finite group, prove that

$$P_\chi = \frac{\dim \chi}{|G|} \sum_{g \in G} \overline{\chi(g)} \rho(g)$$

is the projection onto the χ-isotypic subspace. Apply these projections to the displacement space of a molecule whose equilibrium configuration is preserved by G.

4 Continuous Symmetry and Angular Momentum

Definition 4.1 (Lie groups and Lie algebras)**.** A *Lie group* is a smooth manifold G whose multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_e G$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 4.1 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 4.2 (Infinitesimal representations)**.** For a smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \frac{d}{dt}\Big|_0 \rho(\exp tX).$$

Exercise 4.2 (Differentiating symmetry)**.** Prove that $d\rho$ preserves brackets and that $\rho(\exp X) = \exp(d\rho(X))$. Prove that a connected Lie-group representation is determined by its infinitesimal representation whenever the latter integrates uniquely.

Definition 4.3 (Rotations and spin)**.** Define

$$SO(3) = \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},$$
$$SU(2) = \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}.$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 4.3 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations.

Exercise 4.4 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \ldots\}$, has dimension $2j + 1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j + 1, \ldots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$.

Exercise 4.5 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 4.4 (Quantum symmetries)**.** For a Hilbert space H and self-adjoint Hamiltonian $\mathcal{H}$, a *unitary symmetry* is a unitary U satisfying $U\mathcal{H} = \mathcal{H}U$. A *quantum number* is an eigenvalue of a self-adjoint operator commuting with $\mathcal{H}$. An operator A has *symmetry type* W when its components define an intertwiner from W to $\mathrm{End}(H)$ under conjugation.

Exercise 4.6 (Symmetry, degeneracy, and selection rules)**.** Prove that every eigenspace of $\mathcal{H}$ is symmetry-invariant and decomposes into irreducibles. Deduce symmetry-forced degeneracies. Prove that a transition matrix element $\langle f, Ai\rangle$ can be nonzero only if the trivial representation occurs in $V_f^* \otimes W \otimes V_i$, and derive the angular-momentum rules for an electric-dipole operator.

Exercise 4.7 (Atomic spectra)**.** For the Coulomb Hamiltonian, use rotational symmetry and separation into irreducible $SO(3)$-types to obtain the quantum numbers ℓ, m and their degeneracies. With the radial spectrum $E_n = -m_e e^4/(2(4\pi\varepsilon_0)^2\hbar^2 n^2)$ as derived input, compute the Balmer wavelengths and compare them with the observed hydrogen lines. Apply the electric-dipole rule to identify forbidden transitions.

Exercise 4.8 (Molecular symmetry)**.** For the equilibrium geometry of H_2O, determine the point-group action on nuclear displacements, remove translations and rotations, and decompose the vibrational representation into irreducibles. Determine infrared and Raman activity from the transformation types of the dipole and polarizability. Compare the predicted number and symmetry of lines with the observed three fundamental vibrational bands.

Part III

Fields

5 Manifolds, Forms, and Metrics

Definition 5.1 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth.

Definition 5.2 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$.

Exercise 5.1 (Intrinsic tangent vectors)**.** Prove that derivations at p, equivalence classes of smooth curves through p, and the usual coordinate tangent vectors are naturally isomorphic. Prove the chain rule $d(G \circ F)_p = dG_{F(p)} \circ dF_p$ and compute tangent spaces of regular level sets.

Definition 5.3 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. A *flow* of X is a smooth local action Φ of $\mathbb{R}$ on M satisfying $\frac{d}{dt}\Big|_0 \Phi_t(p) = X_p$. The *Lie bracket* is the vector field $[X, Y](f) = X(Yf) - Y(Xf)$.

Exercise 5.2 (Flows and brackets)**.** Prove local existence and uniqueness of flows. Prove that $[X, Y] = 0$ if and only if their local flows commute, and that the Lie derivative $\mathcal{L}_X Y = [X, Y]$ is natural under diffeomorphisms.

Definition 5.4 (Exterior algebra and differential forms)**.** The *exterior algebra* of a vector space V is $\Lambda V = T(V)/\langle v \otimes v : v \in V\rangle$. A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$. The *pullback* of $\omega \in \Omega^k(N)$ by $F : M \to N$ is

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

Definition 5.5 (Exterior derivative)**.** The *exterior derivative* is the unique family of maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and naturality $F^*d = dF^*$.

Exercise 5.3 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity $\mathcal{L}_X = d\iota_X + \iota_X d$ and the naturality of d, wedge products, and pullbacks.

Definition 5.6 (Integration of forms)**.** An *orientation* of an n-manifold is a continuous choice of component of nonzero elements in $\Lambda^n T_p^*M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 5.4 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem.

Definition 5.7 (Tensor fields and metrics)**.** A (r, s)-*tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. A *Riemannian metric* is a smooth positive-definite inner product on each T_pM. A *Lorentzian metric* is a smooth nondegenerate symmetric form of signature $(-, +, \dots, +)$. The length of a curve in a Riemannian manifold is $\int \sqrt{g(\dot\gamma, \dot\gamma)}\, dt$; the proper time of a timelike curve is $c^{-1} \int \sqrt{-g(\dot\gamma, \dot\gamma)}\, dt$. A metric and an orientation define a volume form vol_g.

Exercise 5.5 (Metric volume and proper time)**.** Construct vol_g intrinsically and derive its coordinate expression. Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, maximize proper time among inertial paths with fixed endpoints and derive special-relativistic time dilation.

6 Connections, Curvature, and Field Equations

Definition 6.1 (Connections and geodesics)**.** A *connection* on TM is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fY) = X(f)Y + f\nabla_X Y$. Its torsion is $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$. A *geodesic* satisfies $\nabla_{\dot\gamma}\dot\gamma = 0$. A connection is *metric compatible* if $Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Exercise 6.1 (Levi–Civita connection)**.** Prove that every Riemannian or Lorentzian metric has a unique torsion-free, metric-compatible connection. Derive the Koszul formula and prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints.

Definition 6.2 (Covariant derivative and parallel transport)**.** For a curve γ, the connection induces a *covariant derivative* D/dt on vector fields along γ. A field V is *parallel* when $DV/dt = 0$; the resulting linear isomorphism between endpoint tangent spaces is *parallel transport*.

Exercise 6.2 (Parallel transport and holonomy)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Show that Levi–Civita transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

Definition 6.3 (Curvature)**.** The *curvature* of a connection is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For a metric connection, the *Riemann tensor* is $\mathrm{Riem}(X, Y, Z, W) = g(R(X, Y)Z, W)$. Its contraction is the *Ricci tensor* Ric; the contraction of Ricci is the *scalar curvature* S.

Exercise 6.3 (Curvature identities and geodesic deviation)**.** Prove tensoriality and the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field J, prove

$$\frac{D^2 J}{dt^2} + R(J, \dot\gamma)\dot\gamma = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

Definition 6.4 (Gauge fields)**.** Let G be a Lie group and $\pi : P \to M$ a principal G-bundle. A *gauge connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. Its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M.

Exercise 6.4 (Gauge covariance and Bianchi identity)**.** Derive the transformation laws for a local gauge potential and its curvature. Prove $d_A F_A = 0$ and gauge invariance of holonomy up to conjugacy. For $G = U(1)$, recover $A \mapsto A + d\chi$ and $F = dA$.

Definition 6.5 (Electromagnetic field)**.** On an oriented Lorentzian four-manifold, an *electromagnetic field strength* is a two-form F. Given a current one-form J and constants μ_0, c, *Maxwell's equations* are

$$dF = 0, \qquad d{*}F = \mu_0 {*}J.$$

Locally, a *gauge potential* is a one-form A with $F = dA$.

Exercise 6.5 (Maxwell theory and electromagnetic waves)**.** Derive charge conservation from Maxwell's equations. On Minkowski spacetime, recover the electric and magnetic vector equations and the Lorentz force from F. In vacuum, impose Lorenz gauge and derive wave equations with speed c. Prove transverse polarization and compare the predicted relation $c = (\mu_0\varepsilon_0)^{-1/2}$ with the measured speed of light.

Definition 6.6 (Einstein equation)**.** The *Einstein tensor* of a Lorentzian metric is $G = \mathrm{Ric} - \frac{1}{2}Sg$. A *stress–energy tensor* is a symmetric two-tensor T whose contraction with an observer's velocity gives the observer's energy–momentum current. The *Einstein equation* with cosmological constant Λ is

$$G + \Lambda g = \frac{8\pi G_N}{c^4} T.$$

Exercise 6.6 (Conservation and the Newtonian limit)**.** Use the contracted Bianchi identity to derive $\nabla^a T_{ab} = 0$. Linearize $g = \eta + h$ for a static weak field and recover Poisson's equation and the Newtonian inverse-square law with the stated coupling constant.

Exercise 6.7 (Gravitational radiation)**.** Linearize the vacuum Einstein equation about Minkowski spacetime, impose transverse–traceless gauge, and obtain two tensor polarizations satisfying a wave equation. Apply geodesic deviation to a ring of freely falling test masses and derive the strain measured by an orthogonal-arm interferometer.

Exercise 6.8 (Gravitational-wave detection)**.** For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. Using public values for GW150914, infer the chirp mass and order of magnitude of the strain. Compare these predictions with the measured coincident LIGO strain, limiting the conclusion to agreement of the waveform with general relativity in the observed regime.

Part IV

States

7 Hamiltonian and Statistical States

Definition 7.1 (Configuration and phase spaces)**.** A *configuration space* is a smooth manifold Q. Its *phase space* is the cotangent bundle T^*Q. The tautological one-form on T^*Q is $\theta_\alpha(v) = \alpha(d\pi(v))$, and its canonical symplectic form is $\omega = -d\theta$.

Definition 7.2 (Symplectic dynamics)**.** A *symplectic form* is a closed, nondegenerate two-form ω. The *Hamiltonian vector field* of $H \in C^\infty(M)$ is defined by $\iota_{X_H}\omega = dH$. Its flow is the *Hamiltonian flow.* The *Poisson bracket* is $\{f, g\} = \omega(X_f, X_g)$.

Exercise 7.1 (Hamilton's equations and Poisson algebra)**.** Prove that the canonical form on T^*Q is symplectic. In canonical coordinates, derive Hamilton's equations. Prove antisymmetry, the Leibniz rule, and the Jacobi identity for the Poisson bracket, and prove $\dot{f} = \{H, f\} + \partial_t f$ along Hamiltonian flow.

Definition 7.3 (Symplectic actions and momentum maps)**.** An action of a Lie group G on (M, ω) is *symplectic* if it preserves ω. A *momentum map* is an equivariant map $J : M \to \mathfrak{g}^*$ such that $d\langle J, \xi\rangle = \iota_{\xi_M}\omega$ for every $\xi \in \mathfrak{g}$.

Exercise 7.2 (Noether's theorem)**.** Prove that if a Hamiltonian is invariant under a symplectic G-action with momentum map J, then every component of J is conserved. Compute the momentum maps for translations and rotations of a particle and recover linear and angular momentum conservation.

Exercise 7.3 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the phase-space volume $\omega^n/n!$, and every density depending only on H. Deduce that Hamiltonian evolution cannot compress phase volume.

Definition 7.4 (Measure and integration)**.** A *σ-algebra* Σ on X is closed under complements and countable unions. A *measure* is a countably additive map $\mu : \Sigma \to [0, \infty]$ with $\mu(\varnothing) = 0$. A function is *measurable* if inverse images of Borel sets are measurable. Its integral is obtained from nonnegative simple functions by monotone limits and then by positive and negative parts. For $1 \le p < \infty$, $L^p(X, \mu)$ consists of measurable functions modulo equality almost everywhere with $\int |f|^p \, d\mu < \infty$; L^∞ uses essential bounds.

Exercise 7.4 (Convergence and L^p spaces)**.** Prove monotone and dominated convergence and Fatou's lemma. Prove Hölder's and Minkowski's inequalities and completeness of L^p. Prove that L^2 is a Hilbert space and identify the dual of L^p for $1 < p < \infty$ under the standard σ-finiteness hypotheses.

Definition 7.5 (Probability)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map. Its *expectation* is its integral. The *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is the $\mathcal{G}$-measurable integrable random variable satisfying $\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X, d\mathbb{P}$ for all $G \in \mathcal{G}$. Families of σ-algebras are *independent* when the probability of every finite intersection is the product of its probabilities.

Exercise 7.5 (Expectation, conditioning, and independence)**.** Prove existence and almost-everywhere uniqueness of conditional expectation, the tower property, and conditional Jensen inequality. Characterize independence by factorization of expectations and derive variance addition for independent square-integrable random variables.

Definition 7.6 (Entropy and Gibbs ensembles)**.** For a discrete probability vector p, its *entropy* is $S(p) = -k_B \sum_i p_i \log p_i$. For a phase-space probability density ρ relative to Liouville measure μ_L, its *statistical entropy* is $S(\rho) = -k_B \int \rho \log \rho, d\mu_L$. The *canonical Gibbs ensemble* at inverse temperature β is $d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H} d\mu_L$.

Definition 7.7 (Measure-preserving dynamics and ergodicity)**.** A measurable transformation T is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$. It is *ergodic* if every invariant measurable set has measure zero or full measure. An *equilibrium state* is an invariant probability measure for the dynamics.

Exercise 7.6 (Entropy and equilibrium)**.** Maximize entropy under normalization and fixed mean energy to derive the Gibbs ensemble. Prove $\langle H \rangle = -\partial_\beta \log Z$ and $\mathrm{Var}(H) = \partial_\beta^2 \log Z$. Prove invariance of Gibbs measure under Hamiltonian flow and formulate the consequence of the ergodic theorem for equality of time and ensemble averages.

Exercise 7.7 (Ideal-gas thermodynamics)**.** Compute the classical canonical partition function of N indistinguishable noninteracting particles in a box, including the phase-space normalization $h^{3N}N!$. Derive the ideal-gas law, internal energy, and entropy, and compare the pressure law with dilute-gas measurements within the classical regime.

8 Hilbert-Space and Algebraic States

Definition 8.1 (Hilbert spaces and bounded operators)**.** A *Hilbert space* is a complete inner-product space. A linear map $T : H \to K$ is *bounded* if $\|T\| = \sup_{\|x\| \le 1} \|Tx\| < \infty$. Its adjoint is the unique bounded map satisfying $\langle Tx, y \rangle = \langle x, T^*y \rangle$. Projections, self-adjoint operators, and unitary operators satisfy respectively $P = P^* = P^2$, $A = A^*$, and $U^*U = UU^* = \mathrm{id}$.

Definition 8.2 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X is a map E from measurable sets to orthogonal projections on H with $E(X) = \mathrm{id}$ and countable additivity in the strong operator topology. For a bounded measurable f, the operator $\int_X f, dE$ is defined by uniform approximation from simple functions.

Exercise 8.1 (Spectral theorem)**.** Prove that every bounded self-adjoint operator A has a unique projection-valued measure on $\mathrm{spec}(A) \subset \mathbb{R}$ such that $A = \int \lambda, dE(\lambda)$. Extend the statement to bounded normal operators, derive continuous functional calculus, and characterize unitary and projection spectra.

Definition 8.3 (Quantum states and observables)**.** A *density operator* is a positive trace-class operator ρ with $\mathrm{Tr}\,\rho = 1$. An *observable* is a self-adjoint operator A with spectral measure E_A. The *Born probability* of an event B is $\mathrm{Tr}(\rho E_A(B))$. A closed-system *unitary dynamics* is $\rho \mapsto U_t \rho U_t^*$ for a strongly continuous unitary group U_t.

Exercise 8.2 (Born rule and unitary dynamics)**.** Prove that the Born rule defines a probability measure and that its expected value is $\mathrm{Tr}(\rho A)$ when defined. Derive the von Neumann equation from $U_t = e^{-itH/\hbar}$. Prove conservation of expected energy and derive the Heisenberg equation for observables.

Definition 8.4 (Composite and reduced states)**.** The state space of a composite system is $H_A \otimes H_B$. A state is *separable* if it lies in the closed convex hull of product density operators and *entangled* otherwise. The *reduced state* $\rho_A = \mathrm{Tr}_B\,\rho$ is determined by $\mathrm{Tr}(\rho_A A) = \mathrm{Tr}(\rho(A \otimes \mathrm{id}))$ for all bounded A.

Exercise 8.3 (Schmidt decomposition and entanglement)**.** Prove the Schmidt decomposition for pure vectors in a tensor product of finite-dimensional Hilbert spaces. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectrum. Compute the reduced states and entanglement entropy of a Bell state.

Definition 8.5 (C^*-algebras and states)**.** A *Banach algebra* is a complete normed algebra with $\|ab\| \le \|a\|\|b\|$. An *involution* satisfies $(ab)^* = b^*a^*$ and $(a^*)^* = a$. A *C^*-algebra* is an involutive Banach algebra satisfying $\|a^*a\| = \|a\|^2$. An element is *positive* if it is b^*b for some b. A *state* is a positive linear functional φ with $\|\varphi\| = \varphi(1) = 1$.

Exercise 8.4 (Commutative Gelfand duality)**.** For compact Hausdorff X, prove that $C(X)$ is a commutative unital C^*-algebra. Prove that every commutative unital C^*-algebra A is isometrically $*$-isomorphic to $C(\operatorname{Spec} A)$ by the Gelfand transform, and identify states on $C(X)$ with probability measures.

Definition 8.6 (Representations and GNS construction)**.** A *representation* of a C^*-algebra A is a $*$-homomorphism $\pi : A \to B(H)$. For a state φ, define $N_\varphi = \{a : \varphi(a^*a) = 0\}$ and $\langle [a], [b] \rangle = \varphi(a^*b)$ on A/N_φ.

Exercise 8.5 (GNS theorem)**.** Complete A/N_φ and prove that left multiplication induces a cyclic representation $(\pi_\varphi, H_\varphi, \Omega_\varphi)$ satisfying $\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle$. Prove uniqueness up to a cyclic unitary equivalence.

Definition 8.7 (CAR algebra and fermionic Fock space)**.** For a complex Hilbert space K, the *canonical anticommutation relations algebra* $\mathrm{CAR}(K)$ is the unital C^*-algebra generated by conjugate-linear symbols $a(f)$ satisfying

$$\{a(f), a(g)\} = 0, \qquad \{a(f), a(g)^*\} = \langle f, g \rangle 1.$$

The *antisymmetric Fock space* is $\mathcal{F}_-(K) = \bigoplus_{n \ge 0} \Lambda^n K$. Its vacuum is $\Omega = 1 \in \Lambda^0 K$; creation by f is exterior multiplication, and annihilation by f is its adjoint.

Exercise 8.6 (CAR and GNS predict fermionic antibunching)**.** Prove that creation and annihilation on $\mathcal{F}_-(K)$ satisfy the CAR and that $\omega_0(x) = \langle \Omega, x\Omega \rangle$ is a state. Apply the preceding exercise to identify its GNS representation with the Fock representation. Let ω_C be a gauge-invariant quasifree state with one-particle density $0 \le C \le 1$, and use its determinantal Wick rule to prove, for orthonormal detector modes f, g, that

$$\omega_C(N_f N_g) = \omega_C(N_f)\omega_C(N_g) - |\langle f, Cg \rangle|^2, \qquad N_f = a(f)^* a(f).$$

Deduce $N_f^2 = N_f$ and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

Definition 8.8 (Operator topologies and von Neumann algebras)**.** The *weak operator topology* is generated by $T \mapsto \langle x, Ty \rangle$; the *strong operator topology* is generated by $T \mapsto \|Tx\|$. For $S \subseteq B(H)$, its *commutant* is $S' = \{T : TS = ST \text{ for all } S \in S\}$. A *von Neumann algebra* is a unital $*$-subalgebra $M \subseteq B(H)$ with $M = M''$.

Exercise 8.7 (Bicommutant theorem)**.** Prove that for a unital $*$-subalgebra $A \subseteq B(H)$, the strong and weak operator closures of A equal A''. Deduce that von Neumann algebras have sufficient projections to approximate their self-adjoint elements spectrally.

Definition 8.9 (Traces, factors, and noncommutative probability)**.** A *trace* on a von Neumann algebra is a normal positive functional τ satisfying $\tau(ab) = \tau(ba)$. A *factor* is a von Neumann algebra with center $\mathbb{C}1$. A *noncommutative probability space* is a pair (M, φ) of a von Neumann algebra and a normal state; its random variables are operators affiliated with M.

Exercise 8.8 (Projection dimension and factors)**.** For a finite factor with normalized trace τ, prove additivity and unitary invariance of τ on projections. Relate the possible projection values to matrix factors and to diffuse finite factors, and formulate independence of commuting subalgebras by factorization of the state.

9 Completely Bounded Structure and Correlations

Definition 9.1 (Positive and completely positive maps)**.** A linear map $\Phi : A \to B$ between C^*-algebras is *positive* if it preserves positive elements and *completely positive* if every $\mathrm{id}_{M_n} \otimes \Phi$ is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a unital completely positive map on observables in the adjoint picture.

Exercise 9.1 (Stinespring and Kraus representations)**.** Prove that a unital completely positive map $\Phi : A \to B(H)$ has a representation $\Phi(a) = V^*\pi(a)V$. In finite dimensions, derive a Kraus representation $\Phi(a) = \sum_i K_i^* a K_i$ and its normalization. Prove that every channel is unitary evolution on a larger system followed by discarding an environment.

Exercise 9.2 (Amplitude damping reproduces spontaneous emission)**.** For a qubit with ground state $|0\rangle$, excited state $|1\rangle$, and environment vacuum $|0\rangle_E$, let

$$\begin{aligned} V_t|0\rangle &= |0\rangle|0\rangle_E, \\ V_t|1\rangle &= \sqrt{\eta_t}\,|1\rangle|0\rangle_E + \sqrt{1-\eta_t}\,|0\rangle|1\rangle_E, \qquad \eta_t = e^{-t/T_1}. \end{aligned}$$

Trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive $\rho_{11}(t) = e^{-t/T_1}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-t/(2T_1)}\rho_{01}(0)$. Compare the predicted decay and the environmental dependence of T_1 with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

Definition 9.2 (Measurements)**.** A *positive operator-valued measure* is a countably additive map E from measurable events to positive operators with $E(X) = 1$. In state ρ, its outcome probability is $\mathrm{Tr}(\rho E(B))$.

Exercise 9.3 (POVMs and dilation)**.** Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions and derive the measurement channel associated to a POVM. Show that state discrimination can require a nonprojective POVM.

Definition 9.3 (Operator spaces)**.** An *operator space* is a closed subspace $E \subseteq B(H)$ with matrix norms inherited from $M_n(E) \subseteq B(H^{\oplus n})$. A map $u : E \to F$ is *completely bounded* if $\|u\|_{cb} = \sup_n \|\,\mathrm{id}_{M_n} \otimes u\| < \infty$. The matrix order of an operator system is the family of positive cones $M_n(E)_+$.

Definition 9.4 (Ancilla-assisted channel discrimination)**.** A *one-use discrimination strategy* for channels $\Phi_1, \ldots, \Phi_m$ prepares a state on $H \otimes K$, applies $\Phi_j \otimes \mathrm{id}_K$ to the unknown channel input, and measures the output. The strategy is *ancilla assisted* when $\dim K > 1$. It is *unambiguous* when every conclusive outcome identifies the channel without error.

Exercise 9.4 (An ancilla improves measurement-channel discrimination)**.** For $0 < \theta < \pi/4$, define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let $\mathcal{M}, \mathcal{N}$ be the two projective measurement channels with bases $\{\phi, \phi^\perp\}$ and $\{\psi, \psi^\perp\}$. Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of $|\phi\rangle$ and $|\psi\rangle$ and prove

$$P_I^{\mathrm{anc}} = \cos(2\theta), \qquad P_I^{\mathrm{sep}} = \frac{1+\cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for $0 < \theta < \pi/4$ and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).

Exercise 9.5 (Ruan's theorem)**.** Prove that concrete operator-space norms satisfy

$$\|x \oplus y\| = \max(\|x\|, \|y\|), \qquad \|axb\| \leq \|a\|\|x\|\|b\|.$$

Prove conversely that every vector space with matrix norms satisfying these axioms admits a complete isometry into $B(H)$.

Exercise 9.6 (Arveson–Wittstock extension)**.** Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing C^*-algebra. Deduce, by the 2×2 operator-system construction, that completely bounded maps admit norm-preserving extensions.

Definition 9.5 (Operator-space tensor products)**.** The *minimal operator-space tensor norm* on $E \otimes F$ is inherited from concrete inclusions in $B(H \otimes K)$. The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \, \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form $u : E \times F \to \mathbb{C}$ is *jointly completely bounded* when its associated matrix amplifications are uniformly bounded.

Exercise 9.7 (Tensor norms linearize bilinear maps)**.** Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

Exercise 9.8 (Grothendieck inequalities)**.** State and prove the finite-dimensional real Grothendieck inequality for bilinear forms on $\ell_\infty^m \times \ell_\infty^n$. Formulate the noncommutative and operator-space Grothendieck inequalities and deduce their dimension-independent comparison of bounded and completely bounded correlations.

Definition 9.6 (Bell functionals and XOR games)**.** A *Bell functional* is an array $M = (M_{xy}^{ab})$ evaluated on conditional distributions by $\langle M, p\rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a, b \mid x, y)$. An *XOR game* is specified by real coefficients M_{xy} and scores correlations $\sum_{x,y} M_{xy}\mathbb{E}[a_x b_y]$ for signs a_x, b_y.

Exercise 9.9 (Classical and quantum correlation bounds)**.** Prove that the classical and quantum XOR values are

$$\sup_{a_x, b_y = \pm 1} \left|\sum M_{xy} a_x b_y\right|, \qquad \sup_{\|u_x\|, \|v_y\| \leq 1} \left|\sum M_{xy} \langle u_x, v_y\rangle\right|.$$

Apply Grothendieck's inequality to bound their ratio. For CHSH, compute the classical bound 2 and quantum bound $2\sqrt{2}$.

Exercise 9.10 (Measured nonclassical correlations)**.** For a polarization-entangled two-photon state, choose four polarization angles attaining the CHSH value $2\sqrt{2}$ and derive the predicted coincidence correlations. Analyze a loophole-free Bell-test data table with its stated uncertainties and determine whether the classical bound is excluded, restricting the conclusion to local hidden-variable models satisfying the experimental assumptions.

Part V

Topology

10 Homotopy, Cohomology, and Bundles

Definition 10.1 (Homotopy and fundamental group)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $H : X \times [0,1] \to Y$ with endpoints f and g. The *fundamental group* $\pi_1(X, x_0)$ is the group of endpoint-fixed homotopy classes of loops based at x_0, with concatenation.

Exercise 10.1 (Fundamental groups and winding)**.** Prove homotopy invariance of π_1 and the Seifert–van Kampen theorem for a union with path-connected intersection. Compute $\pi_1(S^1) \cong \mathbb{Z}$ by lifting loops to $\mathbb{R}$, and prove that the resulting integer is the winding number.

Definition 10.2 (Homology and cohomology)**.** The *singular chain group* $C_n(X; R)$ is the free R-module on continuous maps $\Delta^n \to X$, with alternating face boundary ∂. The *homology* is $H_n(X; R) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. The *cochain complex* is $C^n(X; R) = \operatorname{Hom}(C_n(X; R), R)$ with coboundary $\delta\phi = \phi\partial$, and its cohomology is $H^n(X; R)$.

Definition 10.3 (Products and pairings)**.** The *Kronecker pairing* is $H^n(X; R) \times H_n(X; R) \to R$, $([\phi], [c]) \mapsto \phi(c)$. The *cup product* is the product on cohomology induced by the front–back face formula on cochains.

Exercise 10.2 (Homology, cohomology, and degree)**.** Prove $\partial^2 = 0$, $\delta^2 = 0$, and homotopy invariance. Compute the homology and cohomology rings of spheres and tori. For a continuous map $f : S^n \to S^n$, define its degree by $f_*[S^n] = \deg(f)[S^n]$ and prove multiplicativity and agreement with winding number for $n = 1$.

Definition 10.4 (Fiber, vector, and principal bundles)**.** A *fiber bundle* with fiber F is a map $\pi : E \to X$ locally isomorphic over X to $U \times F \to U$. A *vector bundle* has vector spaces as fibers and linear transition maps. A *principal G-bundle* has a free right G-action and local equivariant trivializations. A *section* is a map $s : X \to E$ with $\pi s = \mathrm{id}_X$.

Exercise 10.3 (Bundles from transition functions)**.** Prove that transition functions satisfying the cocycle condition construct a bundle and that changes of local trivialization give isomorphic bundles. Prove that a principal bundle is trivial exactly when it has a global section. Construct associated vector bundles from representations of G.

Definition 10.5 (Global connections and holonomy)**.** A *connection* on a vector bundle $E \to M$ is a map $\nabla : \Gamma(E) \to \Omega^1(M; E)$ satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Its curvature is $F_\nabla = \nabla^2 \in \Omega^2(M; \operatorname{End} E)$. Its *holonomy* along a loop is the automorphism of the initial fiber given by parallel transport.

Exercise 10.4 (Curvature and holonomy)**.** Derive local connection and curvature forms and their gauge-transformation laws. Prove the Bianchi identity. Show that infinitesimal holonomy is curvature and compute the $U(1)$ holonomy around the boundary of a surface as $\exp(i \int F)$ when a global potential is available.

Definition 10.6 (Chern classes and character)**.** The *Chern classes* of a complex vector bundle E are the natural classes $c_j(E) \in H^{2j}(X; \mathbb{Z})$ determined by $c(E \oplus F) = c(E)c(F)$ and the universal bundle. The *Chern character* is the natural ring map $\mathrm{ch} : K^0(X) \to H^{\mathrm{even}}(X; \mathbb{Q})$ whose value on a line bundle L is $e^{c_1(L)}$.

Exercise 10.5 (Curvature representatives). For a Hermitian bundle with unitary connection, prove that
$$\det\Big(1 + \frac{i}{2\pi}F_\nabla\Big) \quad \text{and} \quad \operatorname{Tr}\exp\Big(\frac{i}{2\pi}F_\nabla\Big)$$
are closed forms whose cohomology classes are independent of the connection. Identify them with the Chern classes and Chern character and prove their Whitney-sum formulas.

Definition 10.7 (Berry geometry). For a smooth family of one-dimensional spectral projections P_x on a Hilbert space, the ranges form a line bundle. The *Berry connection* is $\nabla = P \circ d$ and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral $\langle c_1(L), [\Sigma]\rangle$ is the *Chern number* over an oriented closed surface Σ.

Exercise 10.6 (Berry phase and quantized response). Compute the Berry curvature of a two-level Hamiltonian $H(n) = n \cdot \sigma$ over S^2 and its Chern number. For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as e^2/h times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus.

11 K-Theory and Index

Definition 11.1 (Grothendieck completion and stable equivalence). The *Grothendieck group* of a commutative monoid M is the universal abelian group receiving a monoid map from M. Vector bundles E, F are *stably equivalent* if $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ for some m, n. The group $K^0(X)$ is the Grothendieck group of complex vector bundles on compact X; define $K^1(X) = \widetilde{K}^0(\Sigma X)$.

Exercise 11.1 (Bott periodicity and topological phases). Prove the clutching description of bundles on spheres. Establish Bott periodicity in the form $\widetilde{K}^0(S^{n+2}) \cong \widetilde{K}^0(S^n)$ and compute the complex K-groups of spheres and tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the class-A invariants in dimensions one through three.

Definition 11.2 (Operator K-theory). For a unital C^*-algebra A, $K_0(A)$ is the Grothendieck group of stable homotopy classes of projections in matrix algebras over A, and $K_1(A)$ is the stable homotopy group of unitaries in matrix algebras over A.

Exercise 11.2 (Topological and operator K-theory). Construct the Serre–Swan correspondence between vector bundles over compact X and finitely generated projective $C(X)$-modules. Deduce $K^j(X) \cong K_j(C(X))$ and formulate Bott periodicity for C^*-algebras.

Definition 11.3 (Compact and Fredholm operators). An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator $T : H_0 \to H_1$ is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is $\operatorname{ind} T = \dim\ker T - \dim\operatorname{coker} T$. The *Calkin algebra* is $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$.

Exercise 11.3 (Fredholm index and the Calkin algebra). Prove that T is Fredholm exactly when its image in the Calkin algebra is invertible. Prove homotopy invariance, compact-perturbation invariance, and additivity of the index. Compute the index of the unilateral shift and identify the boundary map $K_1(\mathcal{Q}(H)) \to K_0(\mathcal{K}(H))$.

Definition 11.4 (Elliptic symbols)**.** For vector bundles E, F over a compact smooth manifold, the *principal symbol* of an order-m differential operator $D : \Gamma(E) \to \Gamma(F)$ is the homogeneous bundle map $\sigma_D : \pi^*E \to \pi^*F$ on T^*M. The operator is *elliptic* if $\sigma_D(x, \xi)$ is invertible for every $\xi \neq 0$. Its *analytic index* is its Fredholm index on Sobolev completions; its *topological index* is the K-theoretic pushforward of its symbol class.

Exercise 11.4 (Atiyah–Singer index theorem)**.** Prove elliptic regularity and Fredholmness for elliptic operators on compact manifolds. Establish the Atiyah–Singer equality $\mathrm{ind}_{an} D = \mathrm{ind}_{top}(\sigma_D)$, using Bott periodicity and the axiomatic characterization of the index. Derive the Riemann–Roch and Dirac-index formulas as special cases.

Definition 11.5 (Chirality and instantons)**.** Let M be an oriented even-dimensional Riemannian spin manifold. The *chirality operator* splits its complex spinor bundle as $S = S^+ \oplus S^-$. For a Hermitian vector bundle E with unitary connection A, the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

On an oriented Riemannian four-manifold, an $SU(r)$-connection is an *instanton* or *anti-instanton* when $F_A = *F_A$ or $F_A = -*F_A$. Its *instanton number* is

$$k = \int_M \mathrm{ch}_2(E) = -\frac{1}{8\pi^2} \int_M \mathrm{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

Exercise 11.5 (Fermion chirality detects instanton number)**.** Apply the Dirac-index formula from the preceding exercise to prove

$$\mathrm{ind}\, D_A^+ = \dim\ker D_A^+ - \dim\ker(D_A^+)^* = \int_M \big[\widehat{A}(TM)\,\mathrm{ch}(E)\big]_4.$$

For the fundamental $SU(2)$ bundle over S^4, show that this index is k with the orientation convention in the definition. Use the Weitzenböck formula for a self-dual or anti-self-dual connection to determine which chirality has no zero modes and deduce that the other has exactly $|k|$ zero modes. For a fermion in a representation R, derive the factor $2T(R)k$. Finally, compute the fermionic measure under the axial rotation $\psi \mapsto e^{i\alpha\gamma_5}\psi$ and obtain the integrated anomaly $\Delta Q_5 = 2\,\mathrm{ind}\, D_A^+$, identifying the chirality change carried by an instanton transition.

12 Tensor Categories, Subfactors, and Topological Physics

Definition 12.1 (Monoidal and rigid categories)**.** A *monoidal category* is a category $\mathcal{C}$ with a tensor functor, unit object, and coherent associativity and unit isomorphisms. A *dual* of X is an object X^* with evaluation and coevaluation morphisms satisfying the snake identities. A monoidal category is *rigid* if every object has left and right duals.

Definition 12.2 (Fusion, braided, and modular categories)**.** A *fusion category* over $\mathbb{C}$ is a finite semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit. A *braiding* is a natural family $c_{X,Y} : X \otimes Y \to Y \otimes X$ satisfying the hexagon identities. A ribbon fusion category is *modular* if its braiding is nondegenerate.

Exercise 12.1 (Fusion rules and braiding)**.** Prove that simple-object classes of a fusion category form a based ring under tensor product. Derive the pentagon and hexagon equations for associativity and braiding data. Compute quantum dimensions and braid-group representations for the Fibonacci fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$.

Definition 12.3 (Subfactors and conditional expectations)**.** A *subfactor* is an inclusion $N \subseteq M$ of factors with common unit. A *conditional expectation* $E : M \to N$ is a positive unital N-bimodule projection. For finite factors with normalized trace, the *Jones index* $[M : N]$ is the Murray–von Neumann dimension of $L^2(M)$ as a left N-module.

Definition 12.4 (Basic construction)**.** On $L^2(M)$, the *Jones projection* e_N is the orthogonal projection onto $L^2(N)$. The *basic construction* is $M_1 = \langle M, e_N \rangle''$. Iteration gives the *Jones tower* $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \cdots$.

Exercise 12.2 (Jones tower and Temperley–Lieb relations)**.** For finite-index $N \subseteq M$, derive $e_N x e_N = E(x) e_N$ and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \qquad e_i e_j = e_j e_i \ (|i - j| > 1), \qquad e_i e_{i\pm 1} e_i = [M : N]^{-1} e_i.$$

Exercise 12.3 (Diagrammatic calculus and index quantization)**.** Construct the Temperley–Lieb diagrammatic category, with composition by stacking and each closed loop valued at $\delta = [M : N]^{1/2}$. Use positivity of its Markov trace to prove that index values below 4 are $4\cos^2(\pi/n)$ for integers $n \geq 3$.

Definition 12.5 (Braid groups and Markov traces)**.** The *braid group* B_n has generators σ_i with relations $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ and $\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i - j| > 1$. A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

Exercise 12.4 (Jones polynomial)**.** Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

Definition 12.6 (Chern–Simons theory and Wilson loops)**.** For a connection A on a principal bundle over an oriented three-manifold and an invariant inner product on $\mathfrak{g}$, the *Chern–Simons functional* at level k is

$$S_{CS}(A) = \frac{k}{4\pi} \int_M \left\langle A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right\rangle.$$

For a representation V and closed curve K, the *Wilson loop* is $W_V(K) = \mathrm{Tr}_V(\mathrm{Hol}_A(K))$.

Exercise 12.5 (Gauge invariance, level, and knot invariants)**.** Compute the change of S_{CS} under a gauge transformation and derive the integrality condition on k required for $e^{iS_{CS}}$ to be gauge invariant. Derive flatness as the field equation. Relate Wilson-loop expectations for $SU(2)$ to the Jones polynomial at a root of unity.

Definition 12.7 (Bordisms and topological field theories)**.** The *bordism category* has closed oriented $(n-1)$-manifolds as objects, oriented n-dimensional bordisms as morphisms, and disjoint union as tensor product. An *n-dimensional topological field theory* is a symmetric monoidal functor from this category to vector spaces. An extended theory may assign categories to codimension-two manifolds.

Exercise 12.6 (Functorial gluing)**.** Show that the monoidal-functor axioms give multiplicativity under disjoint union, composition under gluing, duality under orientation reversal, and mapping-class-group representations on state spaces. In two dimensions, prove the equivalence with commutative Frobenius algebras.

Definition 12.8 (Anyons and topological degeneracy)**.** In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

Exercise 12.7 (Abelian and nonabelian anyons)**.** Compute fusion spaces and braid actions for an abelian category and for the Fibonacci category. Prove that nonabelian fusion spaces can have dimension greater than one and that braiding acts projectively on them. Compute the torus degeneracy from the simple anyon types in a modular category.

Exercise 12.8 (Quantum Hall phases and measured topological order)**.** Use flux insertion for a Laughlin state at filling $\nu = 1/m$ to derive quasiparticle charge e/m, exchange phase π/m, Hall conductance $\nu e^2/h$, and genus-g degeneracy m^g. Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.